Online Resource 1: HGFX: reference-compatible hierarchical Gaussian filtering in Python

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Computational Statistics
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S1 Frozen identities and regeneration

The evaluated HGFX release is 1.0.0. Its source commit is

`4dd8fbd8239d05f2c7932a9a9b3b7795f0a9ab27`.

The MATLAB oracle is 8.2.0 at

`2437f4dc241541072722a2695ddeca7b44d83dd3`. These identities define the scope; current development features are not included. Original recovery experiments were not rerun. A new, prospectively bounded released-wheel demonstration uses empirical exchange-rate inputs; it is kept separate from frozen parity evidence.

Candidate figures regenerate with `python generate_figures.py` from this directory. The agreement/accuracy figure is redesigned from the frozen decision; recovery is redrawn in grayscale from the decision, and pyhgf includes a signed residual from all frozen trials. Horizon and backend figures are byte-identical copies of historical images. Input/output hashes and manuscript claims are indexed in the repository ledger. The workflow diagram is conceptual and contains no data. Complete machine-readable source links appear in the repository claim ledger; this supplement supplies a compact reading route.

S2 Fitting and recovery

The official stress fit D02 and holdout D08 retain endpoint/inference failures. In D02, a localized shared-vector objective difference was $2.842170943040401 \times 10^{-14}$; the state discrepancy was zero and the maximum trial-likelihood difference was $8.881784197001252 \times 10^{-16}$. Six of six MATLAB one-spacing start perturbations changed optimizer endpoints. For D08, 13 of 14 such perturbations moved the endpoint outside its unchanged gate. These are exact-scope reference-sensitivity findings, not direct endpoint parity or an explanation for the separate P3 initial-fit failures. The historical M18 scientific failure remains separate from paired S7 accounting.

The paired grid has 72 parameter cases and 36 model-recovery datasets with three candidate fits each. There are 12 shards, two trial lengths (128,256), two truth scales (0.15,0.35 prior SD), six

parameter replicates per stratum and three model replicates per stratum. The selection rule is minimum BIC. Metrics rounded for display; full precision remains in the frozen decision:

Model	Implementation	Convergence	Median r	Median sRMSE
hgf_binary	matlab	0.833333333333	0.202684570158	2.60966284547
hgf_binary	hgfx	0.833333333333	0.202684570141	2.60966284551
ehgf_binary	matlab	0.916666666667	0.461918369101	2.35912688823
ehgf_binary	hgfx	0.916666666667	0.461918369078	2.35912688867
uhgf_binary	matlab	0.791666666667	0.380968313583	2.95788661965
uhgf_binary	hgfx	0.791666666667	0.380968313591	2.95788661807

The frozen scientific criteria are convergence >=0.80, median r >=0.50, median sRMSE <=1.00 and model-recovery balanced accuracy >=0.50. Agreement is 36/36 winners, while balanced accuracy is 0.5833333333333334 in each implementation. These are deterministic grid summaries, not uncertainty intervals.

S2.1 Parameter-wise recovery

The combined transformed-vector indices are 12, 13 and 14 (zero-based): omega2, omega3 and log zeta. Each row below pools 24 cases, including returned nonconverged fits. Values come from the original aggregate archive, whose SHA-256 matches the digest in the retained S7 decision. No new fitting was performed.

Model	Parameter	r MATLAB / HGFX	sRMSE MATLAB / HGFX
hgf	omega2	0.988404 / 0.988404	0.155871 / 0.155871
hgf	omega3	-0.396490 / -0.396490	2.771273 / 2.771273
hgf	log zeta	0.202685 / 0.202685	2.609663 / 2.609663
ehgf	omega2	0.732432 / 0.732432	1.006879 / 1.006879
ehgf	omega3	0.145726 / 0.145726	2.398578 / 2.398578
ehgf	log zeta	0.461918 / 0.461918	2.359127 / 2.359127
uhgf	omega2	0.970916 / 0.970916	0.245734 / 0.245734
uhgf	omega3	0.198098 / 0.198098	3.170042 / 3.170042
uhgf	log zeta	0.380968 / 0.380968	2.957887 / 2.957887

The per-parameter scale is the population standard deviation (ddof=0) of its 24 generated transformed truths, not the prior standard deviation or a standard error. sRMSE divides RMSE by this scale, except for the historical fallback scale of one if the standard deviation is at most 10^{-12}. The median correlation excludes nonfinite correlations; the median sRMSE includes all three parameters. These deterministic perturbation-grid summaries are not population confidence intervals. The grid's truth rule is a sinusoidal perturbation of transformed prior means at two scales, not an independent random population sample.

S2.2 Model recovery confusion matrix

Both implementations have the same row-normalized matrix. Rows are generating models; columns are selected models. There are 12 datasets per generating model. Counts below are recovered exactly from the original proportions and denominator.

Generating model	Selected HGF	Selected eHGF	Selected uHGF
HGF	4/12	2/12	6/12
eHGF	0/12	11/12	1/12
uHGF	4/12	2/12	6/12

Balanced accuracy is $(4/12 + 11/12 + 6/12)/3 = 0.5833333333333334$. The eHGF diagonal differs substantially from the other two, which a single accuracy value conceals. Source: evidence/s7_aggregate.json; the archived ZIP is retained so access does not depend on an Actions artifact remaining online.

S3 Trial-horizon integrity and retrospective localization

The original run 35272347167 contains 24 shards, 144 parameter cases and 72 model cases. Its frozen gate remains false. The 24 original shard archives were reacquired and verified against GitHub archive digests. The original checker, aggregator and diagnostic sources are retained from execution commit `a051e747a3a60054159c81a8b8405fb4b2398277`. The new audit only classifies existing records; it does not modify that run or re-estimate parameters.

Recorded outcome	Cases	Interpretation
Matching nonempty model selections	62 of 72	Same selected model
No selected model in either implementation	10 of 72	Invalid generation, not different winners
Two different nonempty selected models	0 of 72	None observed; original full-grid gate still fails
Shared invalid parameter-case simulations	20 of 144	No fitted parameter pair
HGFX initial rejection / MATLAB return	4 of 144	Asymmetric execution; see below
Successful parameter-fit pairs	120 of 144	All recorded free-index vectors match the expected vector

The checker defines winner_match as equality AND a nonempty selection. Its free-index test compares returned vectors with the expected vector even when a fit did not run or failed and returned an empty vector. Thus 24 index-contract failures comprise the 20 shared invalid simulations and four asymmetric cases. They do not demonstrate reordering among successful pairs. The aggregator uses global coverage and winner conditions for each family's diagnostic label; the three mismatch labels are not three independently localized model defects.

The four asymmetric cases all use classic HGF, truth scale 0.35, at the two longer horizons. HGFX records an unstable initial objective before entering optimization. MATLAB marks a function return successful even when its objective is nonfinite. The recorded cases are:

Trials	Replicate (zero-based)	MATLAB negative log joint	HGFX stage
512	0	negative infinity	Initial objective rejected
512	4	247.25740970068986	Initial objective rejected
1024	0	negative infinity	Initial objective rejected
1024	4	620.4345219314272	Initial objective rejected

The control-flow boundary is localized to the initial-objective admissibility check in the recovery fitting routine, not to a demonstrated optimizer endpoint competition. Historical records do not contain the paired intermediate states needed to identify the first divergent trial, level and variable, or to decide whether the two finite MATLAB cases reflect a semantic or numerical difference. No root-cause correction or universal fit-equivalence claim is made. The long-horizon grid stays excluded from the supported paired-fitting claim; reproducing or repairing it requires a separately frozen diagnostic exercise.

The retained generator errors report invalid variational approximation. The ten absent winners are counted as unsuccessful cases, not discarded. Row-normalized confusion matrices in the historical P3 aggregate normalize selectable models only; their accuracy must not be read as all-case success. This is another reason not to use those diagnostic accuracies as an identifiability result.

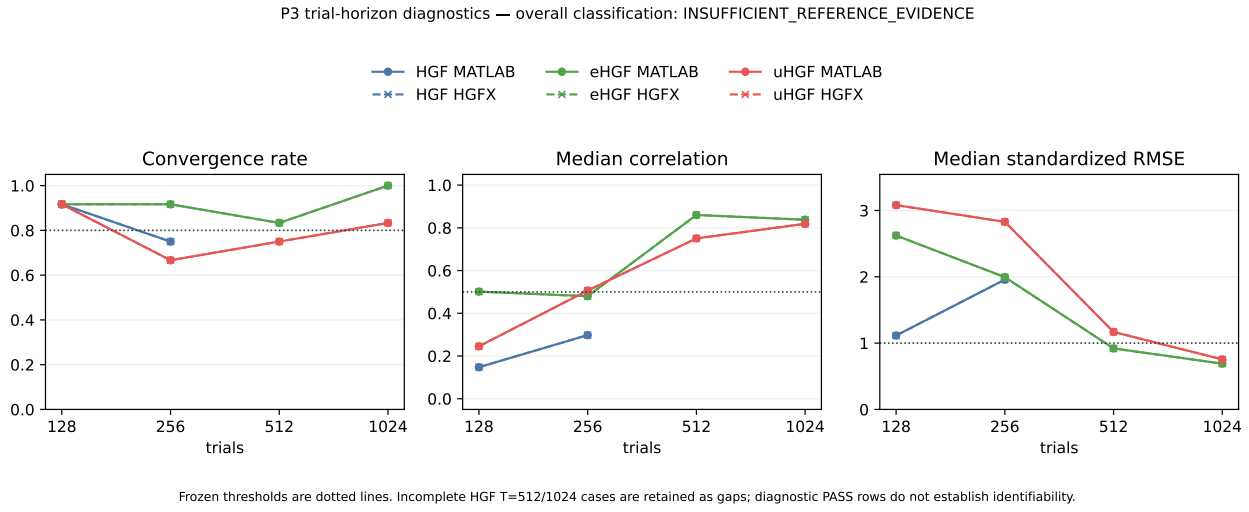


Fig. S1 Historical diagnostic recovery metrics by horizon, retained without changing values. Dotted lines are frozen thresholds and gaps preserve invalid classic-HGF cases. The full-grid integrity gate fails. The retrospective audit distinguishes missing outputs from conflicting finite selections; it does not promote any plotted subset to a scientific success

S4 Backend applicability

Physical evidence is retained from source `07b45a569e04e8e71244c5310dd2cc53dbb2b0ec`, Python 3.12.13, JAX/JAXLIB 0.11.1. Two Tesla T4 devices were visible; four required fitting cells ran on cuda:0. Residency checks and enumeration are separate. Its raw JSON and retention basis remain

archived; it is not a new post-remediation GPU execution. The CPU remeasurement has its own source/runtime identity. The maximum gaps are 0.006783711260709424 (compatibility/JAX CPU, criterion 0.10) and 1.4210854715202004e-14 (JAX CPU/T4, criterion 1e-7).

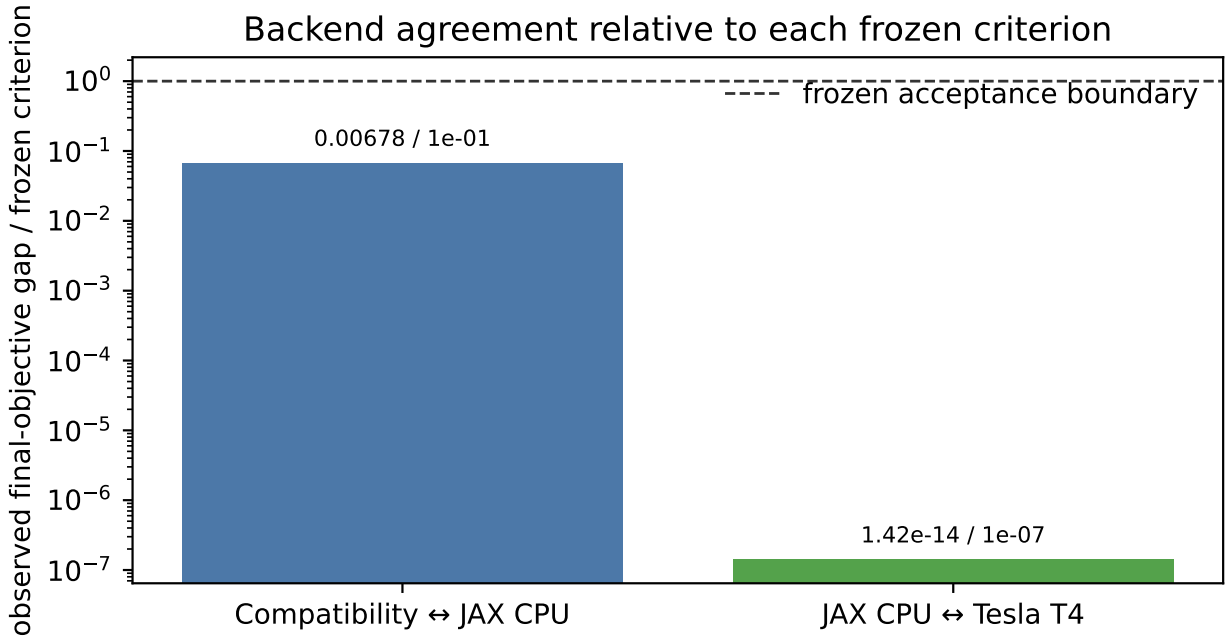


Fig. S2 Each maximum objective gap divided by its own frozen criterion; the dashed line is the acceptance boundary. These bars describe numerical agreement in recorded cells, without error bars. There is no speed measurement

S5 pyhgf common scope

pyhgf 0.3.2 source is `ccd43db5ee5abe4a5a35077d098e53cce2c070c2`. The fixed 128-trial case uses a three-level binary hierarchy, standard volatility updates and mean-field updates, zero drift and unit coupling. The native response inverse temperature is 48. Input and response hashes and the prospective semantic mapping remain in the original common-scope record. Field-wise measured errors:

Quantity	Classification	Maximum absolute error
derived first level input surprise	PASS	4.440892098500626e-16
derived first level prediction error	PASS	1.1102230246251565e-16
first level predicted probability	PASS	1.1102230246251565e-16
level 2 posterior mean	PASS	4.440892098500626e-16
level 2 posterior precision	PASS	6.661338147750939e-16
level 2 predicted mean	PASS	4.440892098500626e-16
level 2 predicted precision	PASS	6.661338147750939e-16
level 3 posterior mean	PASS	1.1102230246251565e-16
level 3 posterior precision	PASS	1.5543122344752192e-15
level 3 predicted mean	PASS	1.1102230246251565e-16
level 3 predicted precision	PASS	1.5543122344752192e-15
participant response nll per trial	NDC	not applicable
participant response nll total	NDC	not applicable

NDC applies to participant-response NLL at a probability boundary; belief agreement does not repair the response comparison. No clipping or changed tolerance was introduced after results. Package-wide or fitted-model equivalence is untested.

S6 Empirical example and published-study boundaries

The input file is `demo/example_usdchf.txt` from the frozen HGF Toolbox 8.2.0 commit. The upstream demo identifies it as USD/CHF rates during 2010/2011. The series has 614 finite values. Its original market-data provider and individual calendar labels are not supplied in this example, so the plots use observation number and do not claim independent market-data verification. The included copy retains the toolbox's MIT license and notice.

Protocol `REVISION_PROTOCOL.md` was committed as `a25690a` before execution. An isolated hgfx==1.0.0 wheel environment ran the whole series outside the checkout with Python 3.12.10, JAX 0.11.1, NumPy 2.5.3, CPU and x64 enabled. The call uses empty responses, hgf_config, bayes_optimal_config and quasinewton_optim_config, matching the official continuous-input workflow. The likelihood is $\log p(u_t) = -\frac{1}{2}\log(2\pi\hat{\sigma}^2_{1,t}) - (u_t - \hat{\mu}_{1,t})^2/(2\hat{\sigma}^2_{1,t})$. No response parameters are fitted. Defaults resolve input-dependent priors from the first observation and first 20 inputs; they are not hand-selected here.

The returned negative log likelihood is -2387.6219521878747, negative log joint -2378.7708383887175, BIC -4736.723934806867 and approximate LME 2376.7070907350367. Execution took 182.001 seconds on this workstation. These are descriptive values from one execution, not a speed benchmark or comparison with a null model. Ridders numerical differentiation emitted invalid-subtraction warnings during the run. The public result did not expose a termination-reason field; convergence and Hessian quality are therefore not certified by the finite returned statistics. No replacement optimizer, repeated fit or alternative prior was tried.

Full transformed parameters, all 614 predicted/posterior state rows, package identity, input digest and command are retained in empirical_example/. Figure 5 uses the observed series and those saved states. Fitting used the full series, so predictions are conditional in-sample diagnostics, not a held-out test. No independent MATLAB comparison was executed for this new example.

The earlier 24-trial synthetic smoke test remains available in worked_example/. Its wheel acquisition used a mirror after the primary host failed DNS resolution; wheel bytes were checked against the official PyPI hash. The empirical run reuses that isolated environment rather than importing the active development checkout.

Pain source commit is `3f1d32aff73826a1385c359ca638ad0e11a7fda7`. Its input audit resolved the apparent filename discrepancy, but custom-response permission and the historical environment remain open. Some simulation files have GPL notices; no unlicensed custom source was translated into HGFX. SPIRL source commit is `86fd5122a744acfb5bc7ba12720426d8b4aa0ffa`; its custom eHGF and seven joint choice/RT models require separate migration. Its 399 random starts and original MATLAB R2019b settings were not silently simplified. Neither study was fitted. The exchange-rate example adds empirical-input use but does not close external participant-study replication. Upstream participant files remain local and ignored.

S7 Published behavioral input validation

The comparison used the two source files below. The preparation code verified their SHA-256 digests before deriving the exact source-defined block inventory.

- Suthaharan dataset (Suthaharan et al. 2021), 130 participant prefixes and 260 blocks. Pinned source file.
- Reed dataset (Reed et al. 2020), 32 participant prefixes and 64 condition halves. Pinned source file.

The initial source gate rejected the assumed 326-block inventory before full-cohort filtering. The MATLAB variable `ans` was outside the participant schema. The corrected protocol fixed the 324-block population before running the paired scientific computation. It retained all observed source blocks and applied the union of missing-choice and missing-outcome indices as the ignored-trial mask.

The perceptual model was `hgf_ar1_binary_mab`, with three arms, uncoupled updates and regular intervals. The native vector was $(\mathrm{NaN}, 0, 1, \mathrm{NaN}, 0.1, 1, \mathrm{NaN}, 0, 0.2, \mathrm{NaN}, 0, 1, 1, 1, \mathrm{NaN}, -2, -6)$. The workflow used MATLAB R2025b Update 7 and the HGF Toolbox 8.2.0 source identified in S1, and the released HGFX 1.0.0 wheel with CPU binary64. It compared complete trajectories and inference states with the inherited tolerances of 2×10^{-9} relative and 2×10^{-11} absolute. Matched nonfinite entries were included in the array-entry count.

The executed run retained all 324 comparison outcomes and exported an aggregate and status artifact. The protocol, source digests, environment and result are recorded with the reproduction code. The summary counts 48 blocks with Python warnings. Participant inputs and per-trial outputs were excluded from the public artifact.

The archived summary was retrieved again during the 8 November editorial revision. Its ZIP SHA-256 was verified as `1adfd1cd69b6f7d7a89ba2c8b9d564e774fd8a8841fa7243d6972605bd9f7391`. The unchanged JSON member, retained at `evidence/published_input_41809919116.json`, has SHA-256 `47432c35aa4914a652a8f1dc2aad81dc4d76fb5175dd0fc1fd47532b81756b70`. This retrieval verifies the aggregate record. The underlying per-trial comparisons were performed in the recorded workflow.

S8. Bounded diagnostic and application follow-up (8 November 2026)

Recounting the archived 324-block status records found 86 warning events in 48 Reed blocks: 29 blocks had one event and 19 had three. All 48 blocks included ignored trials. This association does not establish a warning mechanism. The retained aggregate omits warning messages, categories and source locations, and contains no trialwise arrays with which to reassess nonfinite masks. No fresh MATLAB or HGFX execution is claimed here. The original fixed-vector comparison remains unchanged; harmlessness of the warnings is not established.

A bounded source feasibility check considered Oberwelland before Gao, without fitting either. Oberwelland's public fitting script uses the softmax_binary observation configuration, which is outside the released HGFX 1.0.0 public fitting dispatch. Its OSF file metadata returned HTTP 410 in this attempt. Gao's OSF file metadata returned HTTP 401; critical source files could not be inspected. Access failures are session observations, not evidence of permanent source unavailability. Neither candidate qualified for the conditional application phase; models were not substituted and participants were not selected by results. These findings are source-audit outcomes, not additional participant-study replications.

Machine-readable count and source-access records accompany the repository handoff as publication_completion_audit_20261108.json and publication_completion_sources_20261108.json under the candidate evidence directory. Existing scientific results and acceptance criteria are preserved.

References

Reed EJ, Uddenberg S, Suthaharan P, et al (2020) Paranoia as a deficit in non-social belief updating. eLife 9:e56345. https://doi.org/10.7554/eLife.56345

Suthaharan P, Reed EJ, Leptourgos P, et al (2021) Paranoia and belief updating during the COVID-19 crisis. Nature Human Behaviour 5:1190–1202. https://doi.org/10.1038/s41562-021-01176-8